

Tech Nova Software Hackathon

Project Report

Project: Smart Mall Vehicle Parking Management System

Hackathon: Tech Nova Software Hackathon

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1. Abstract

This project is a smart parking management system for malls that enables customers to view parking availability, reserve slots, generate QR codes, and allows security personnel to verify entry using QR scanning. Administrators monitor parking usage and analytics.

2. Problem Statement

Traditional parking systems waste time and cause congestion due to manual slot management.

3. Objectives

Provide real-time parking availability, QR-based entry, secure role-based access, and efficient parking management.

4. Proposed Solution

A full-stack web application with customer, security, and admin modules.

5. Technologies Used

Frontend: React, Vite, TypeScript, Tailwind CSS

Backend: Node.js, Express.js

Database: PostgreSQL

Authentication: JWT

Real-time: Socket.IO

Version Control: Git & GitHub

6. Modules

Authentication, Customer Dashboard, Booking & QR Code, Security Dashboard, Admin Dashboard, Notifications, Booking History.

7. Database

Tables include Users, ParkingSlots, Bookings, Vehicles, EntryLogs, ExitLogs, Notifications.

8. Workflow

User registers, books slot, QR generated, security scans QR, slot becomes occupied, exit frees slot.

9. Future Enhancements

Online payments, AI-based prediction, license plate recognition, mobile app.

10. Conclusion

The system improves parking efficiency, reduces waiting time, and provides secure access management.